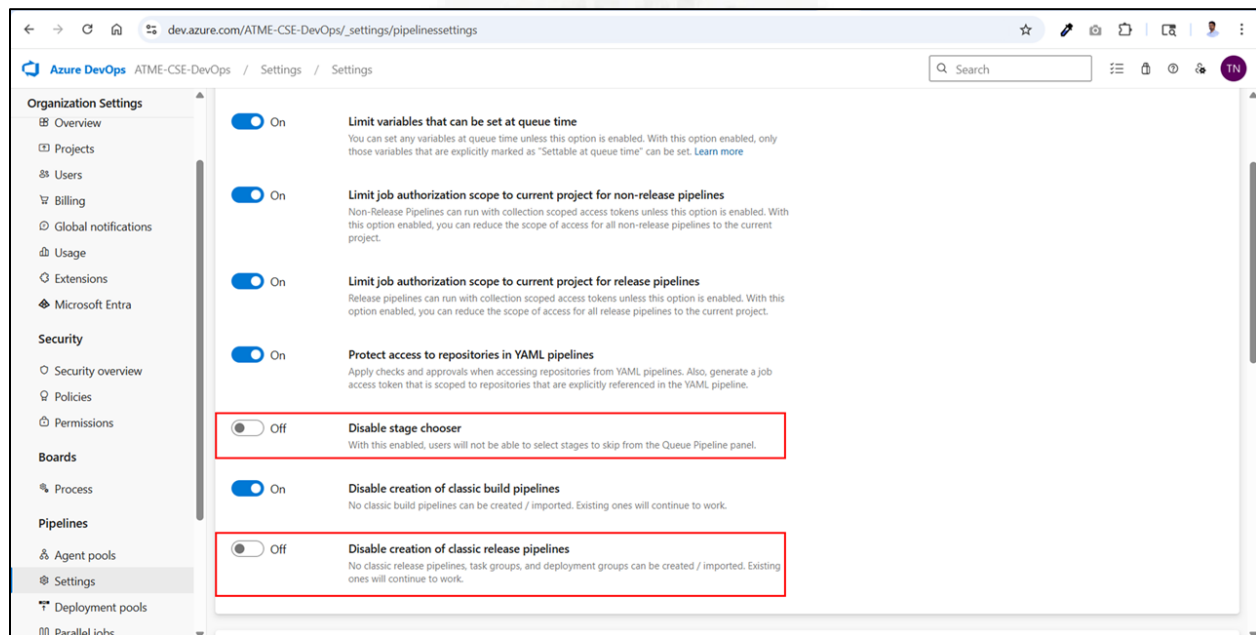


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DevOps Lab Experiment – 11

Creating Release Pipelines: Deploying Applications to Azure App Services, Managing Secrets and Configuration with Azure Key Vault, Hands-On: Continuous Deployment with Azure Pipelines.

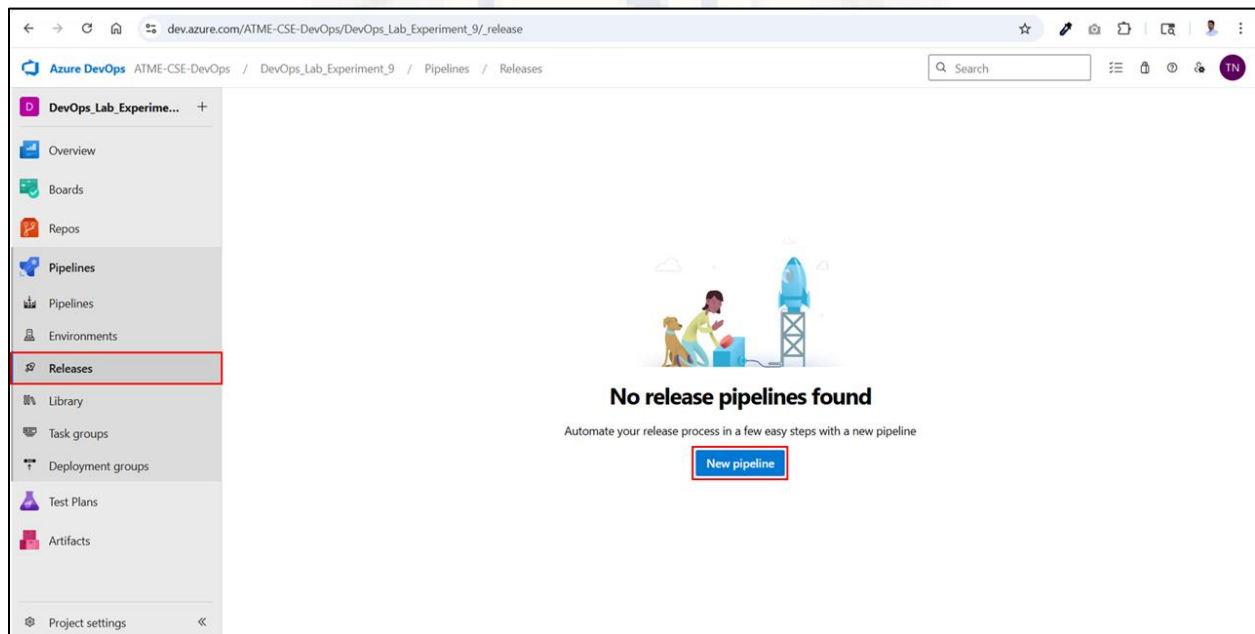
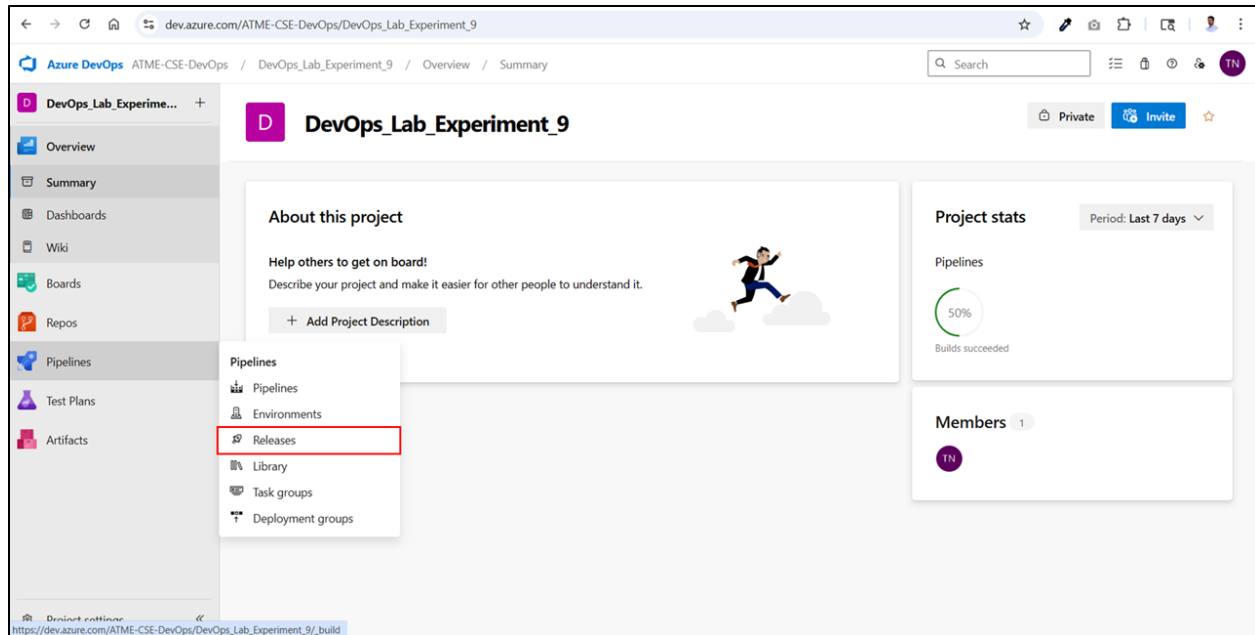
Step 1: Ensure that required settings are enabled under **Organization Settings** to enforce secure and modern pipeline practices.



Step 2 : Create a release pipeline

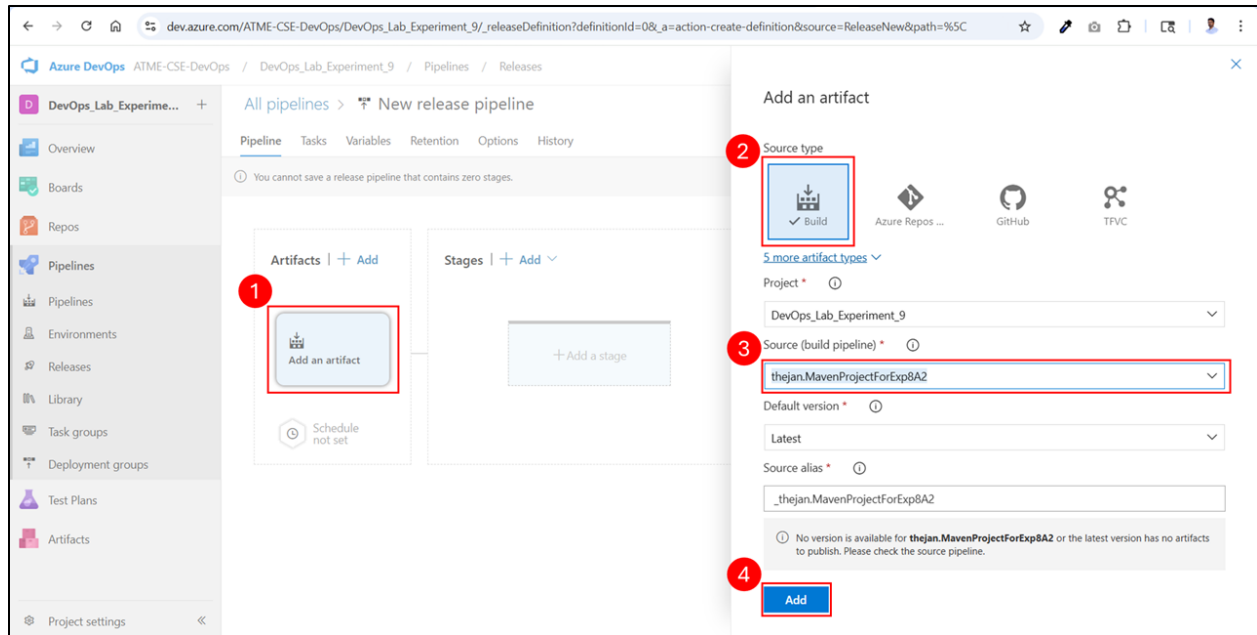
- Open your project → Click on Pipelines → Releases → New pipeline

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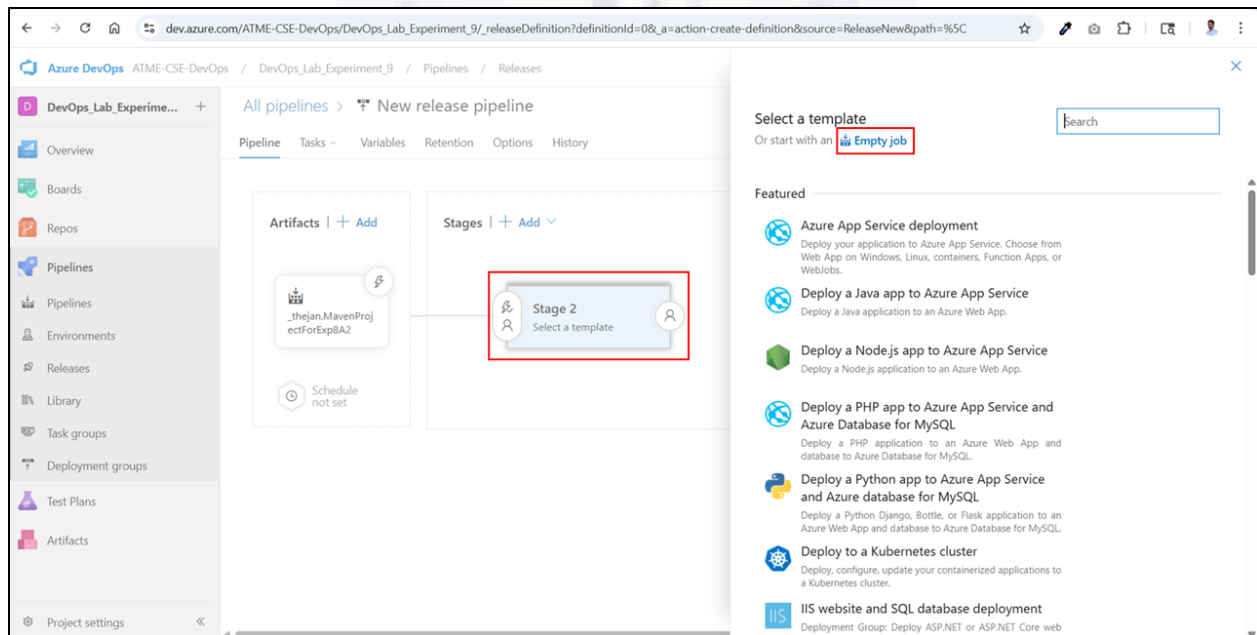
Step 3 : Begin the release pipeline configuration by adding an **artifact** and defining at least one **stage** and select the **build pipeline** as the artifact source to link your build output to the release pipeline

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Adding an artifact

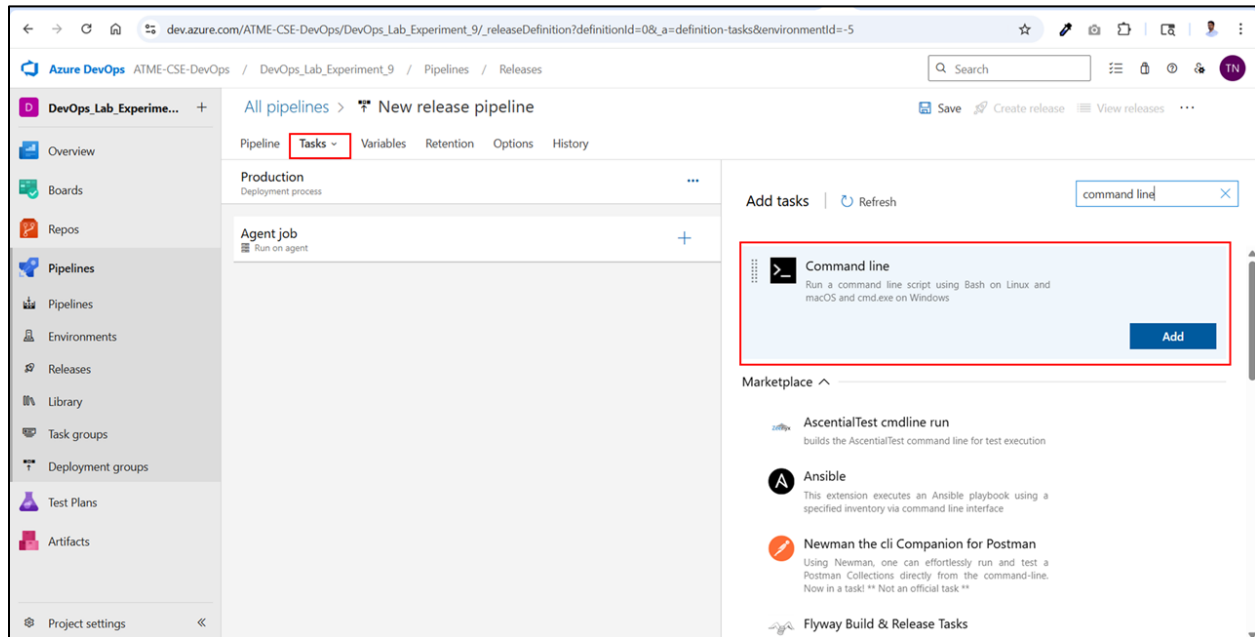
Step 4 : Add a stage and choose a deployment template or start with an **empty job**. Rename the stage if required (Ex. Production)



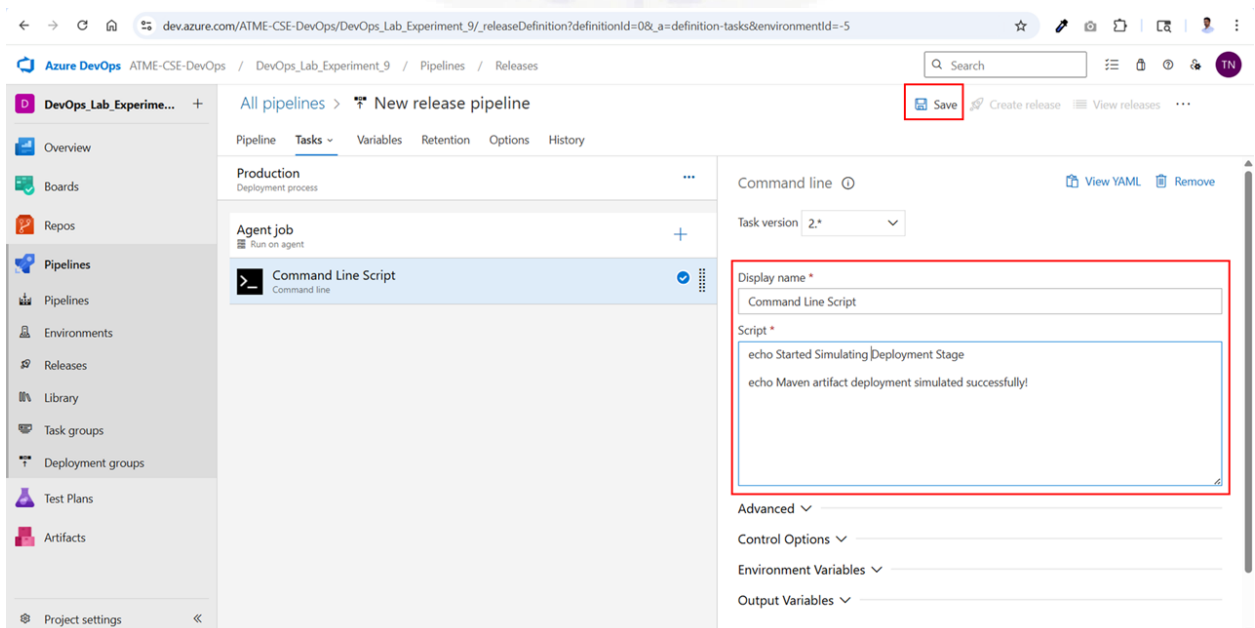
Adding a stage

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Step 5 : Once the stage is created, select the stage to configure jobs and tasks for deployment. Under Tasks, add a task to the agent job, such as **Command Line**, to execute deployment scripts.



Step 6 : In the command line task, enter deployment commands to simulate deployment or perform deployment actions and click on Save.

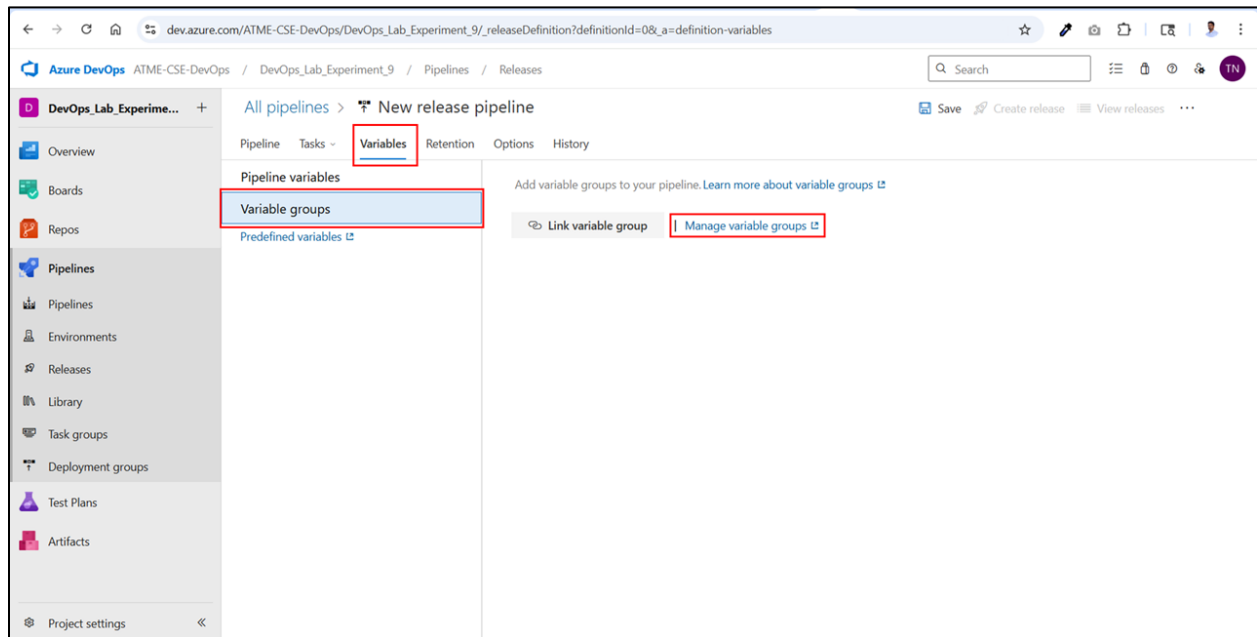


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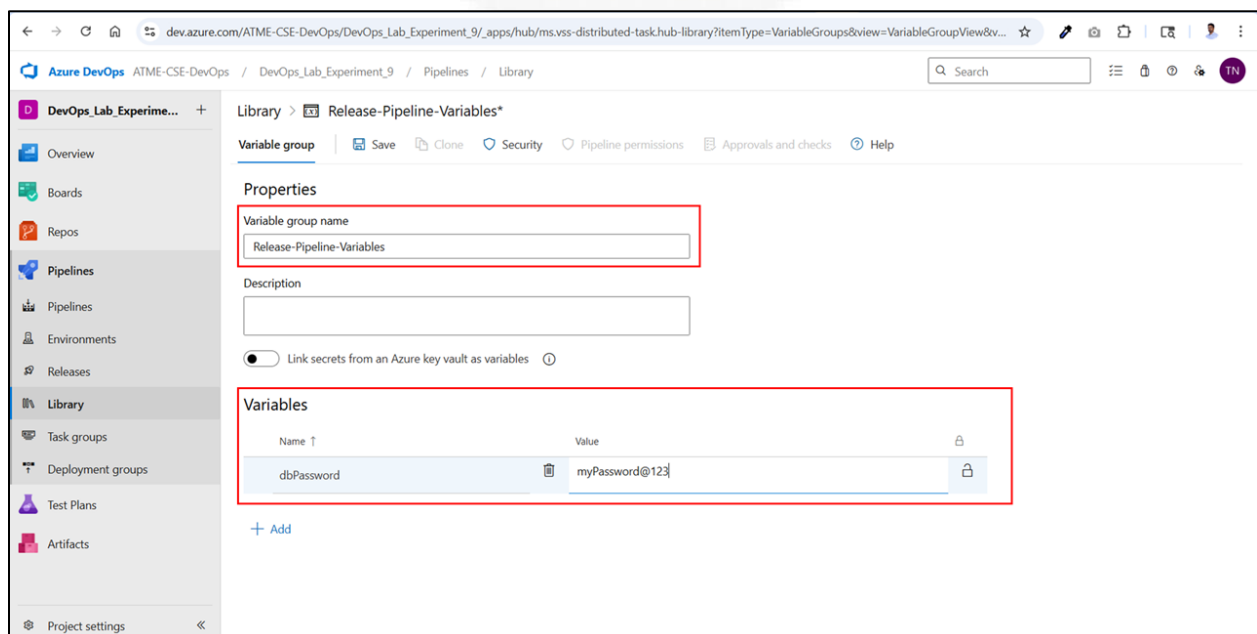
In the command line task, we are only simulating the deployment process using echo commands. No actual deployment is performed; the script simply displays messages indicating the execution of the deployment stage.

Step 7 : Simulate Secrets (Key Vault)

- Under Variables → Select Variables groups → Manage variable groups



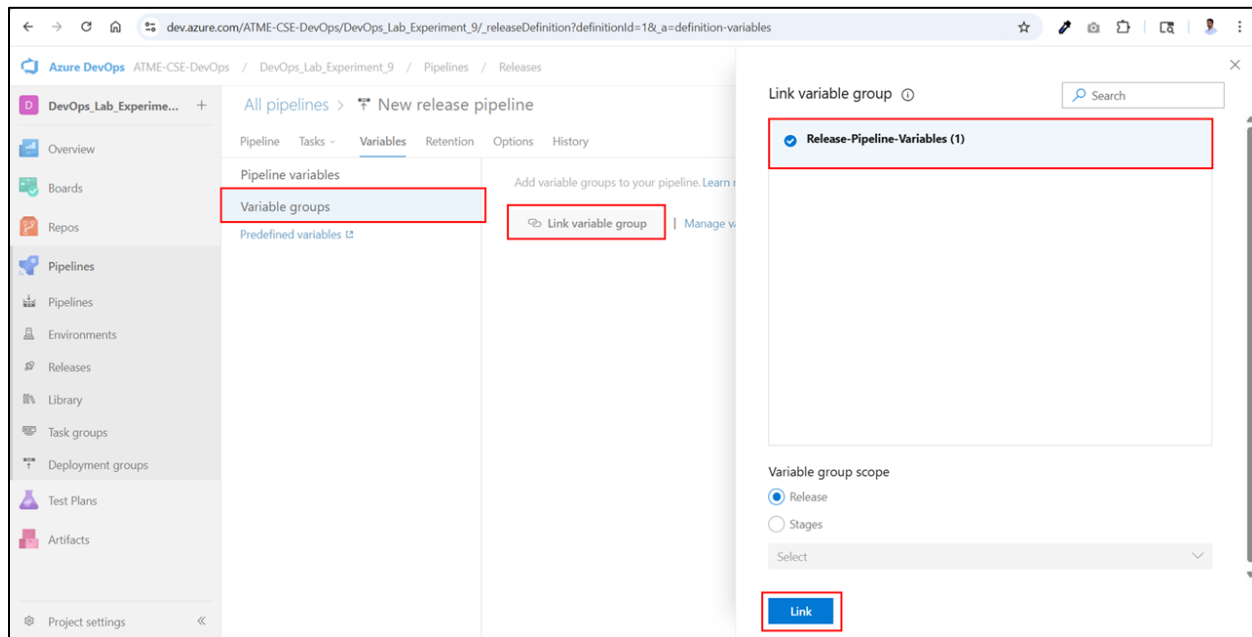
- Provide a name for the Variable group. Add variable by specifying its name and its value



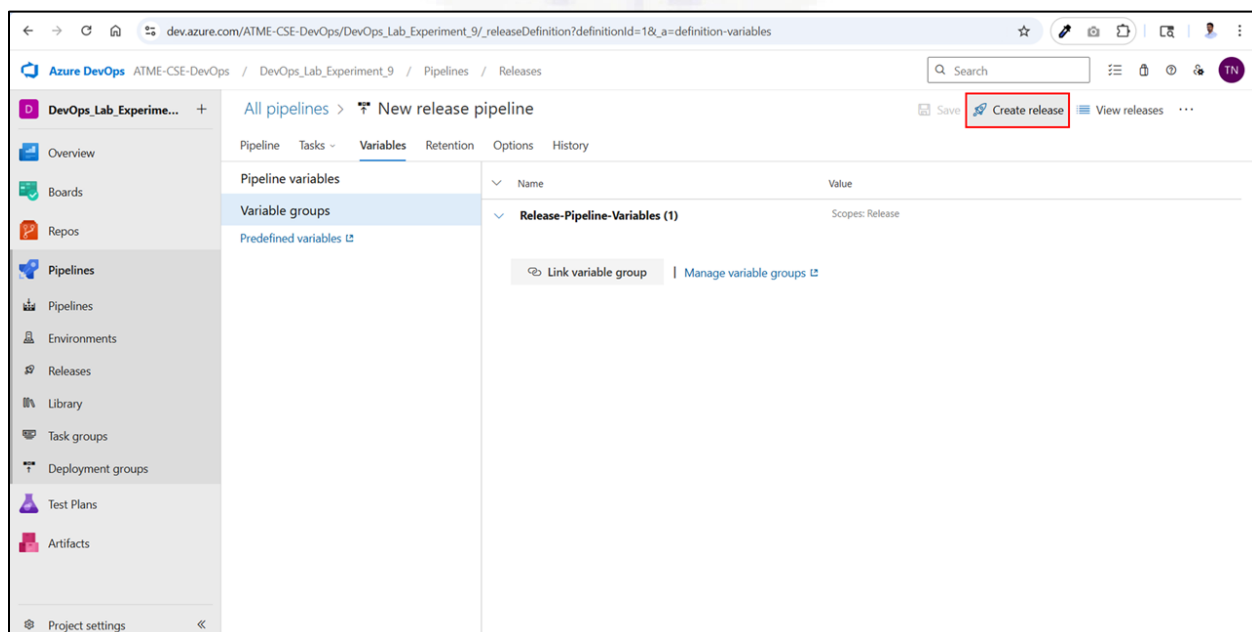
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Step 8: Link the Variable group in the release pipeline

Under Variables → Variable groups → click on Link variable group → select the required Variable group → Link to associate it with the pipeline.

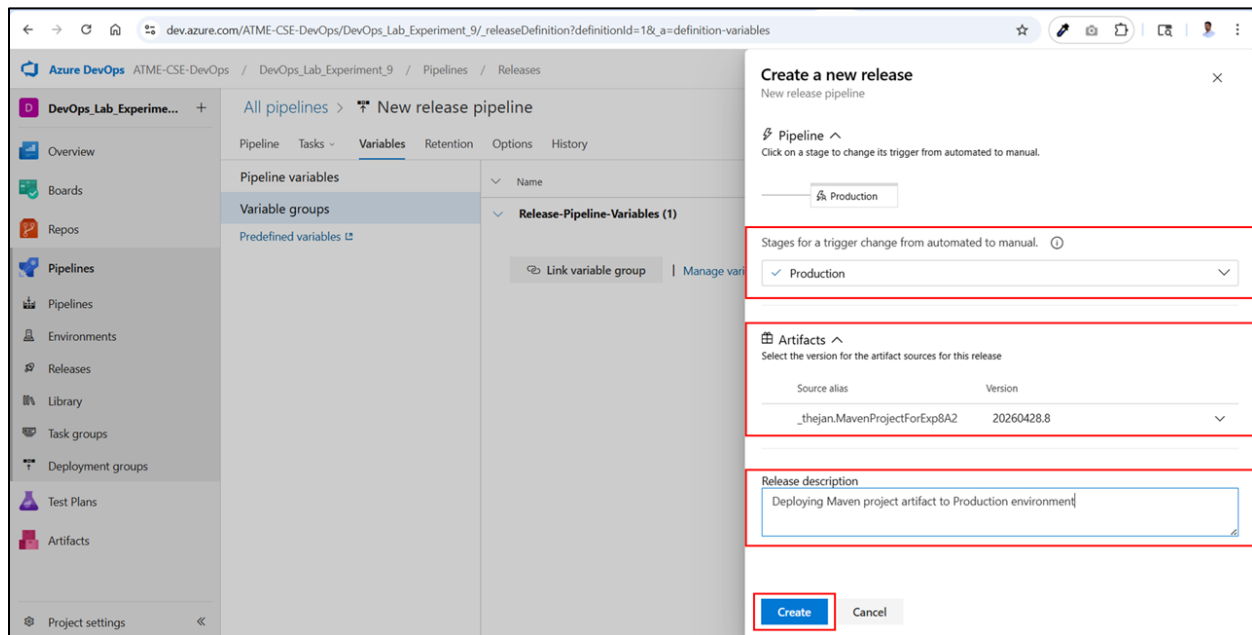


Step 9 : Click on Create a release to trigger the release pipeline manually



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Select the stage which runs during the release, build output for deployment and enter the release description.



dev.azure.com/ATME-CSE-DevOps/DevOps_Lab_Experiment_9/_releaseDefinition?definitionId=1&a=definition-variables

Azure DevOps ATME-CSE-DevOps / DevOps_Lab_Experiment_9 / Pipelines / Releases

DevOps_Lab_Experiment_9 +

All pipelines > New release pipeline

Pipeline Tasks Variables Retention Options History

Pipeline variables

Variable groups

Predefined variables

Release-Pipeline-Variables (1)

Link variable group | Manage variables

Create a new release

New release pipeline

Pipeline

Click on a stage to change its trigger from automated to manual.

Production

Stages for a trigger change from automated to manual.

Production

Artifacts

Select the version for the artifact sources for this release

Source alias Version

_thejan.MavenProjectForExp8A2 20260428.8

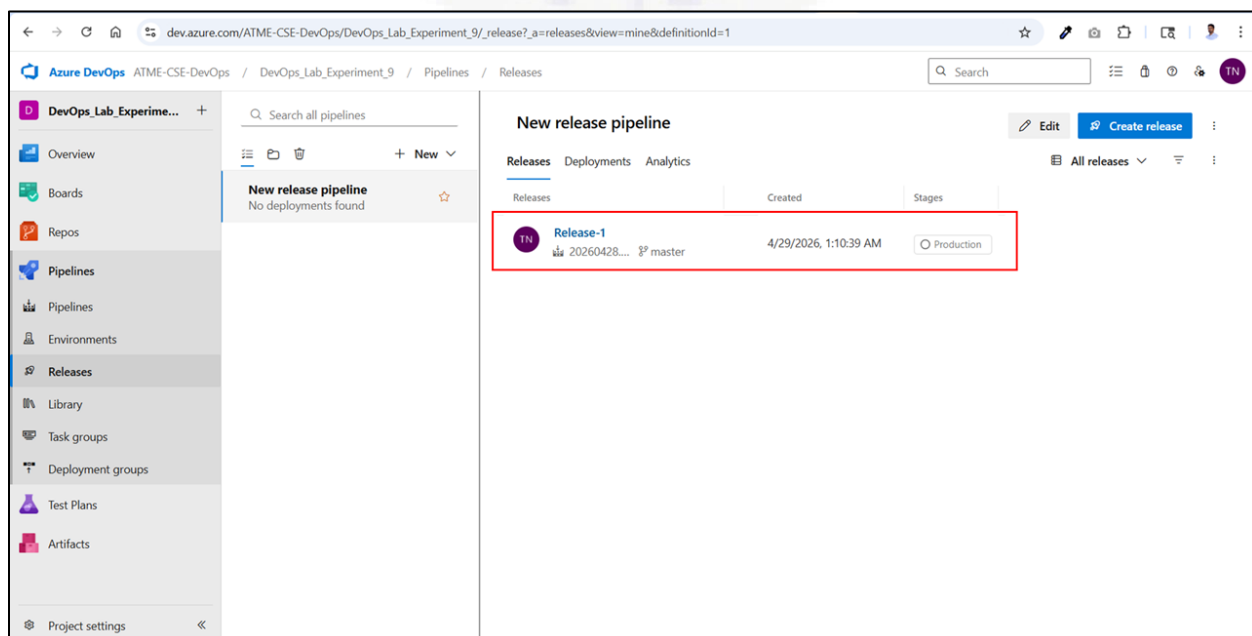
Release description

Deploying Maven project artifact to Production environment

Create Cancel

Step 10 : Run the release pipeline

- Select the release, the stage and click on Deploy



dev.azure.com/ATME-CSE-DevOps/DevOps_Lab_Experiment_9/_release?_a=releases&view=mine&definitionId=1

Azure DevOps ATME-CSE-DevOps / DevOps_Lab_Experiment_9 / Pipelines / Releases

DevOps_Lab_Experiment_9 +

Search all pipelines

New release pipeline

No deployments found

New

Releases Deployments Analytics

Releases Created Stages

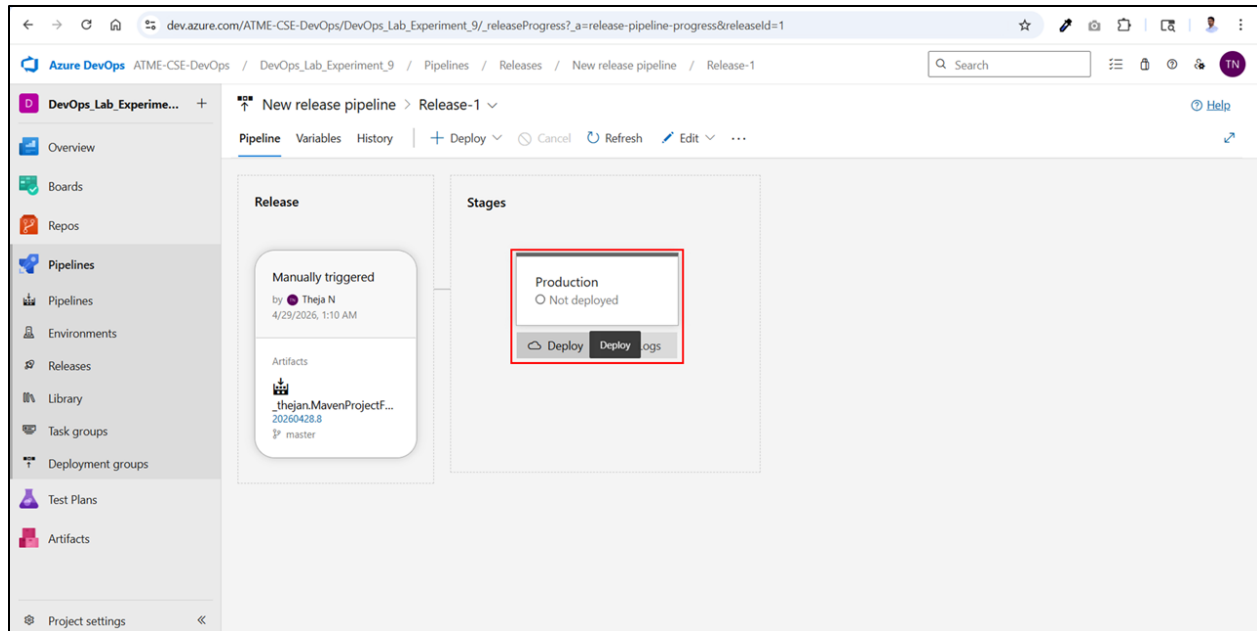
Release-1

20260428.8 master

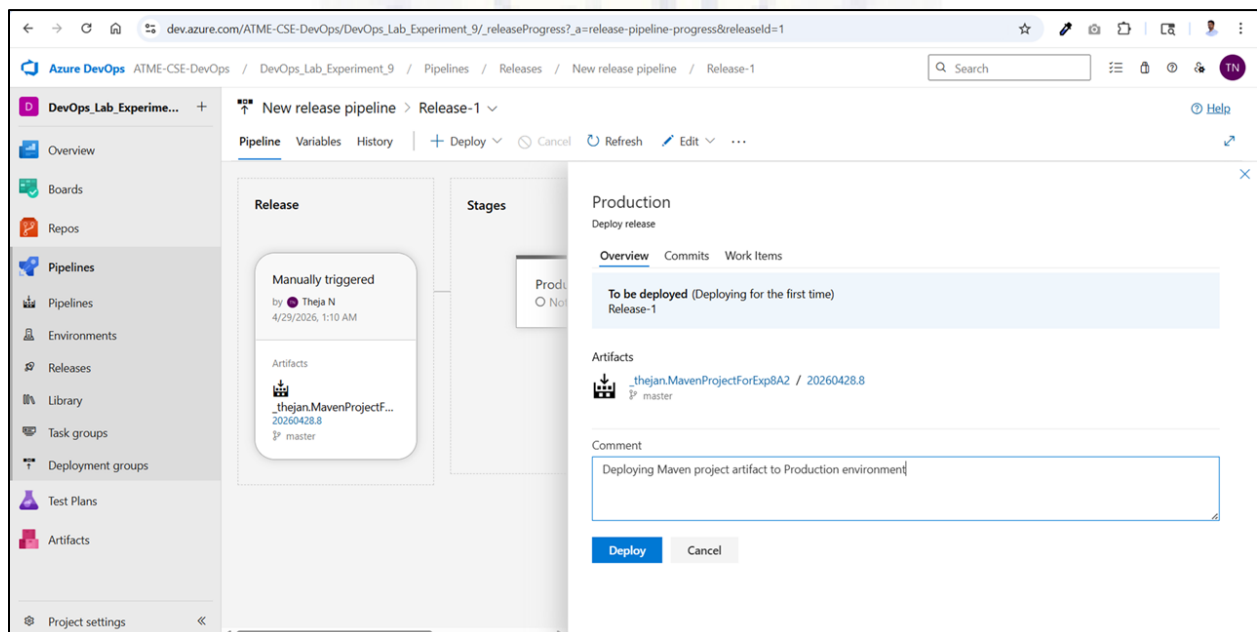
4/29/2026, 1:10:39 AM

Production

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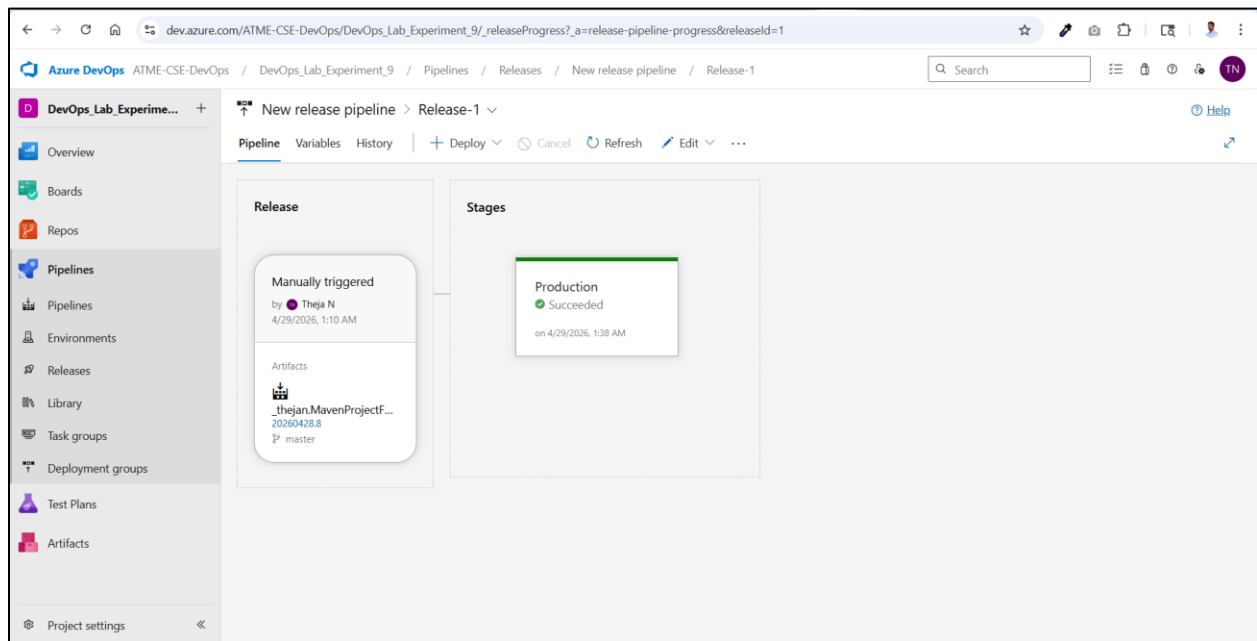


- In the deploy release window, verify the selected artifact, enter a comment and click Deploy



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After clicking **Deploy**, the release pipeline is executed, and the status of the deployment can be viewed in the **Release Pipeline dashboard**. The stage (e.g., *Production*) will show **Succeeded** if the deployment is completed successfully.



If a self-hosted agent is configured on the local machine, the job execution can be observed directly on the agent. This is achieved by selecting the appropriate agent pool (e.g., Default) for the agent job.

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VIVA QUESTIONS

1. What is a release pipeline in Azure DevOps?
2. What is the difference between a build pipeline and a release pipeline?
3. What is meant by Continuous Deployment (CD)?
4. What is an artifact and why is it used in a release pipeline?
5. What is a stage in a release pipeline?
6. What is the role of an agent in Azure DevOps?
7. What is the difference between a Microsoft-hosted agent and a self-hosted agent?
8. What is an agent pool and why is the “Default” pool used?
9. What is a Command Line task and why is it used in this experiment?
10. Why are echo commands used in the deployment step?
11. What are variable groups in Azure DevOps?
12. Why do we use variable groups instead of hardcoding values?
13. How do you link a variable group to a release pipeline?
14. What happens when you click “Deploy” in a release pipeline?
15. How do you verify that the deployment was successful?